

PAPER_664: UQFF White Hole Stability

Subtitle: 4-proof framework demonstrating UQFF extends white hole lifetime by $\sim 30\times$, making Sgr A*-mass white holes cosmologically stable. **Module:** White-HoleStabilityUQFF

Session: Session 172

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Status: Complete — CP4 #248 | UQFF Session 172

Abstract

White holes are classically unstable—they evaporate immediately upon formation. We present four mathematical proofs demonstrating that UQFF vacuum fields collectively extend white hole lifetimes by a factor ~ 30.2 , rendering Sgr A*-scale white holes effectively eternal.

Proof 1 — f_TRZ Negentropic Confinement

$$L' = L_{\text{WH}} \cdot (1 - f_{\text{TRZ}}) \quad \tau' = \tau_{\text{std}} / (1 - f_{\text{TRZ}}) \approx 1.11 \times \tau_{\text{std}} [+11\%]$$

Proof 2 — [UA]/[SCm] Density Gradient

$$L'' = L' \cdot |1 - \rho_{UA}/\rho_{SCm}|^{-1}$$
$$\tau'' = \tau' \cdot |1 - \rho_{UA}/\rho_{SCm}| \approx 10 \tau'$$

Proof 3 — U_m Magnetic String Anchoring

$$L_{UQFF} = L'' \cdot \exp\left(-\frac{U_m}{k_B |T_{WH}|}\right)$$
$$\tau_{UQFF} = \tau'' \cdot \exp\left(\frac{U_m}{k_B |T_{WH}|}\right) \approx 2.718 \tau''$$

Proof 4 — Combined Result

$$\tau_{UQFF} = \frac{\tau_{std}}{1 - f_{TRZ}} \cdot \left|1 - \frac{\rho_{UA}}{\rho_{SCm}}\right| \cdot \exp\left(\frac{U_m}{k_B |T_{WH}|}\right)$$

Combined factor: $1.11 \times 10 \times 2.718 \approx 30.2\times$

Sgr A* (M = 8.55×10^{36} kg): $\tau_{\text{WH,UQFF}} \square$ Hubble time $\rightarrow$ **effectively eternal.**

C++ Module

WhiteHoleStabilityUQFF.h / .cpp — Session 172 CP4 #248 — WhiteHoleStabilityUQFFCalculator

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